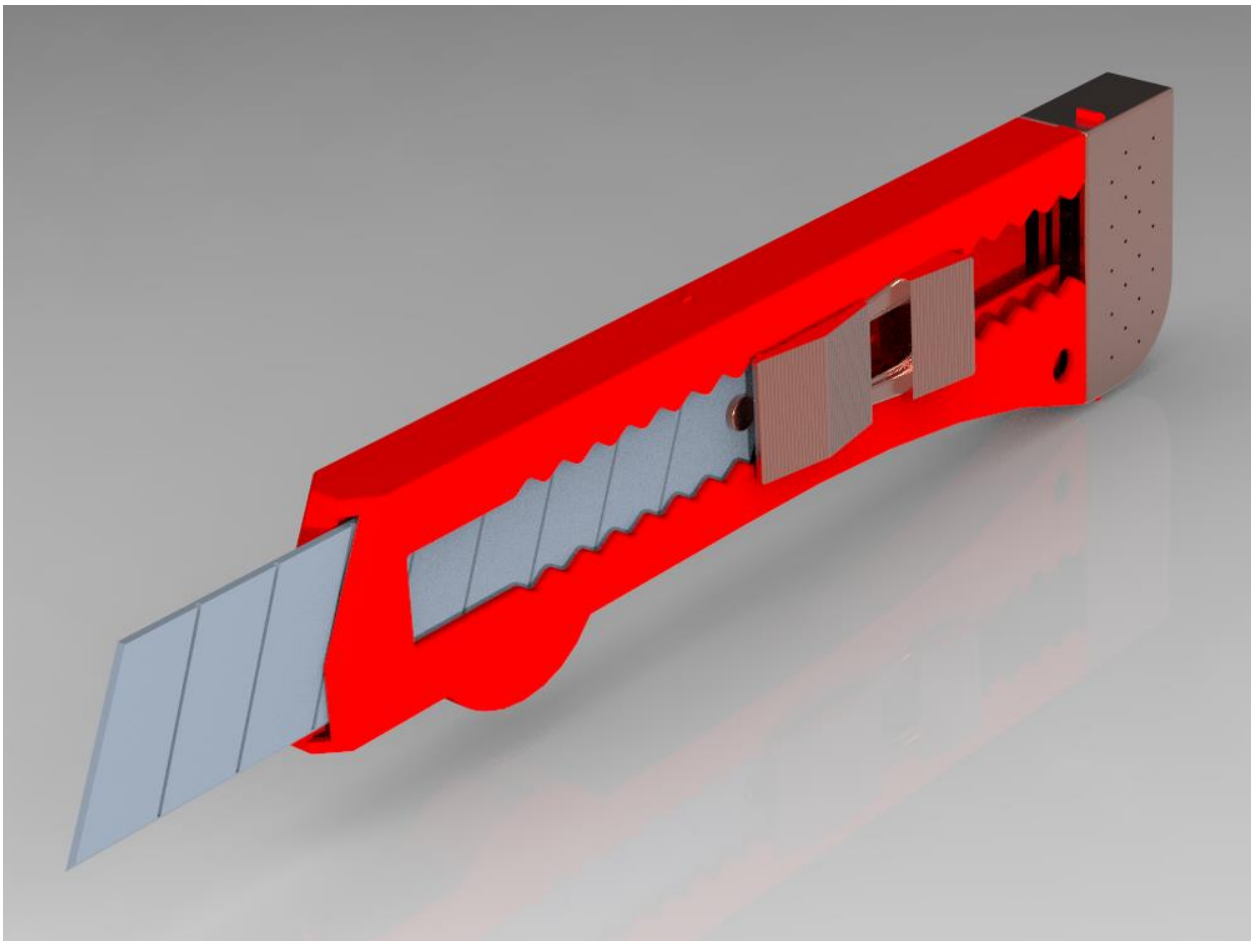
Mechanical Engineering Design Portfolio

By Sarkis Kazandjian (MSME)

Retractable Knife Design

Role:
Freelance mechanical engineer who led the design.

Project goal:
Use advanced CAD modelling techniques to design retractable utility knives with a given set of specifications. Perform advanced injection molding simulations and CAM processes to undertake 3 or 5 axis CNC machining operations.

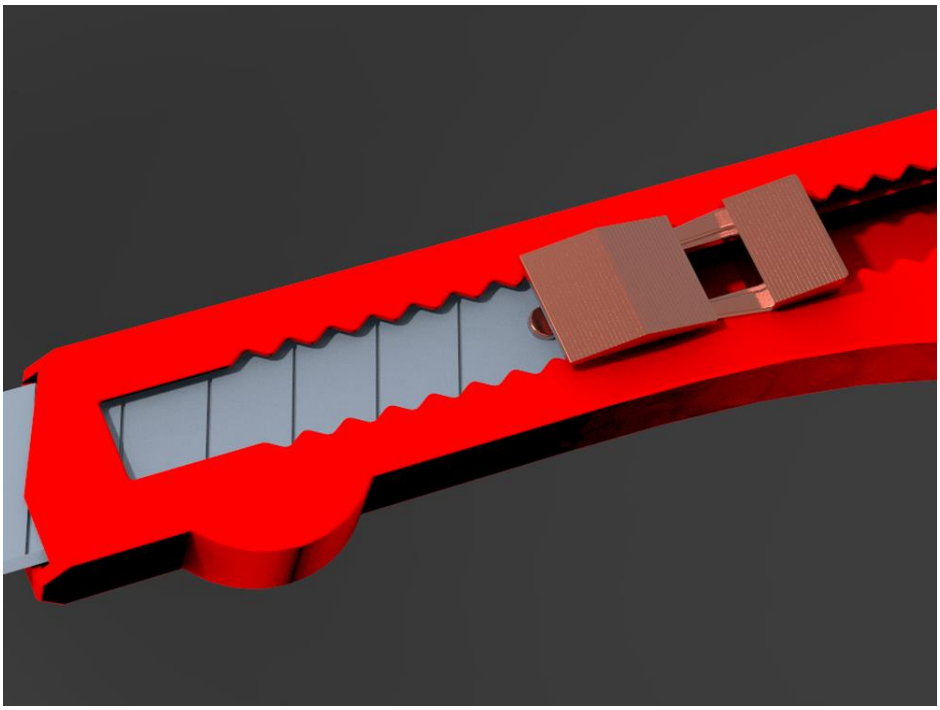


Isometric View of Retractable Utility Knife

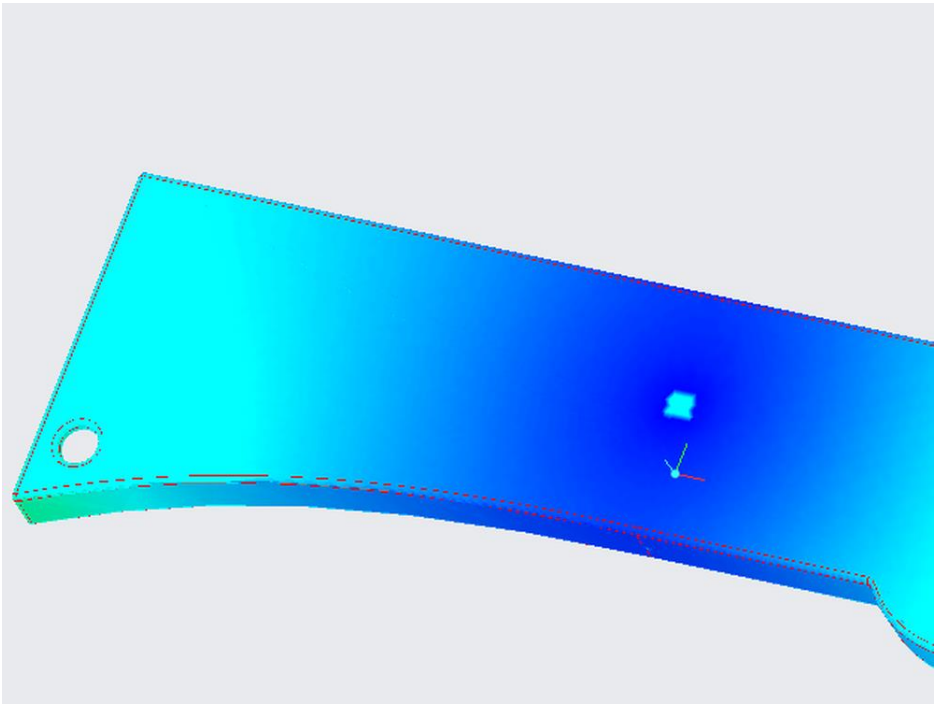
Mechanical Engineering Services:
CAE, 3D Modeling, CAD, CAM, FEA, CFD

Mechanical Engineering Software:
CREO

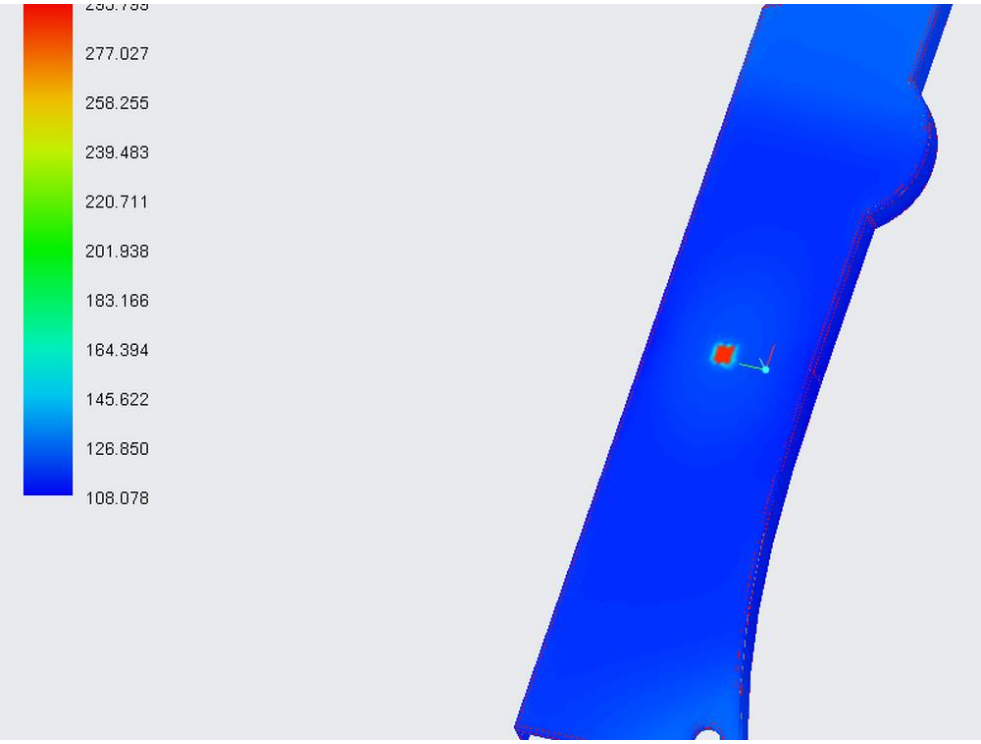
Solution:



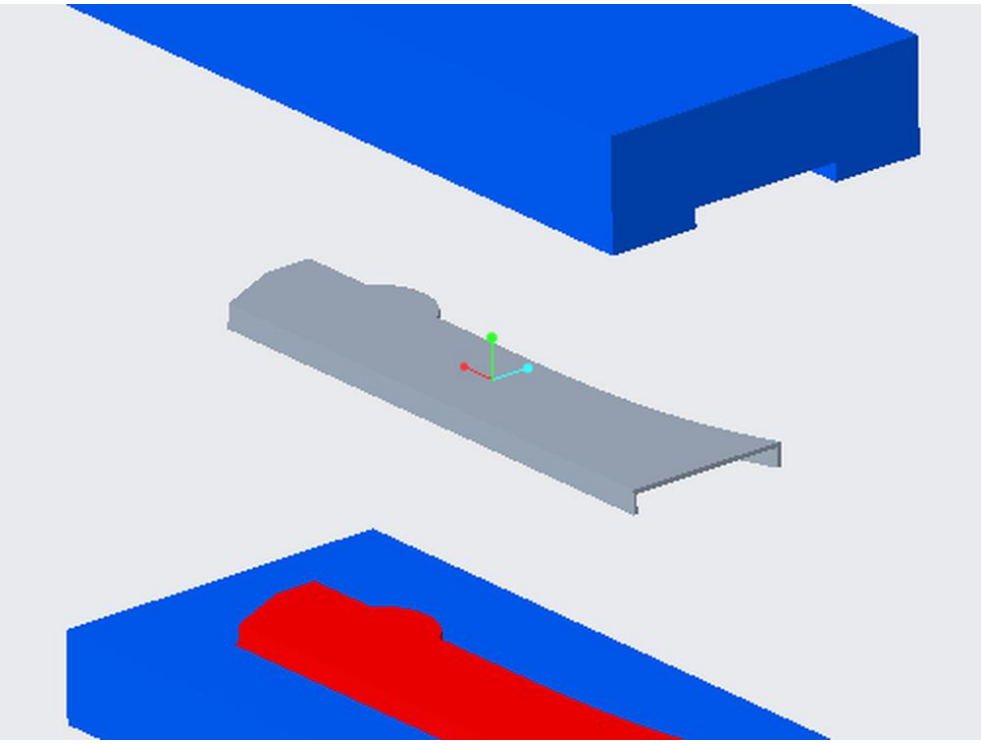
Front View of Retractable Utility Knife



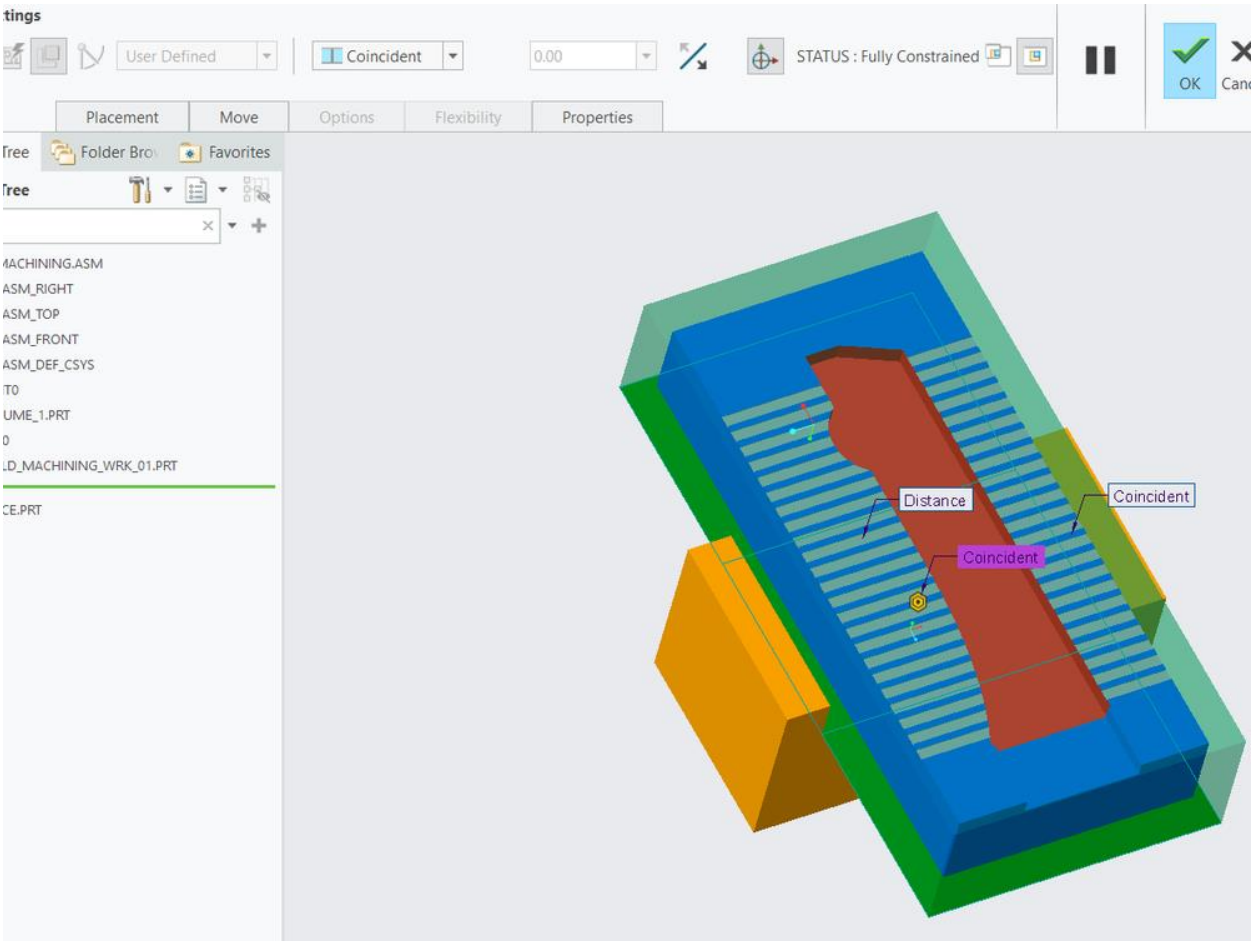
Volume Shrinkage Percentage



Mold Temperature Distribution



Mold Opening with Cavity and Core



Mold and Vice Assembly for Machining Operations

Finite Element Analysis (FEA) of Bicycle Crank Subjected to Impact Loading

Role:
Graduate research student who performed the analysis.

Project goal:
A real-life commercial bicycle crank failed after sustaining fatigue and impact loading. The aim was to simulate the impact failure using FEA.

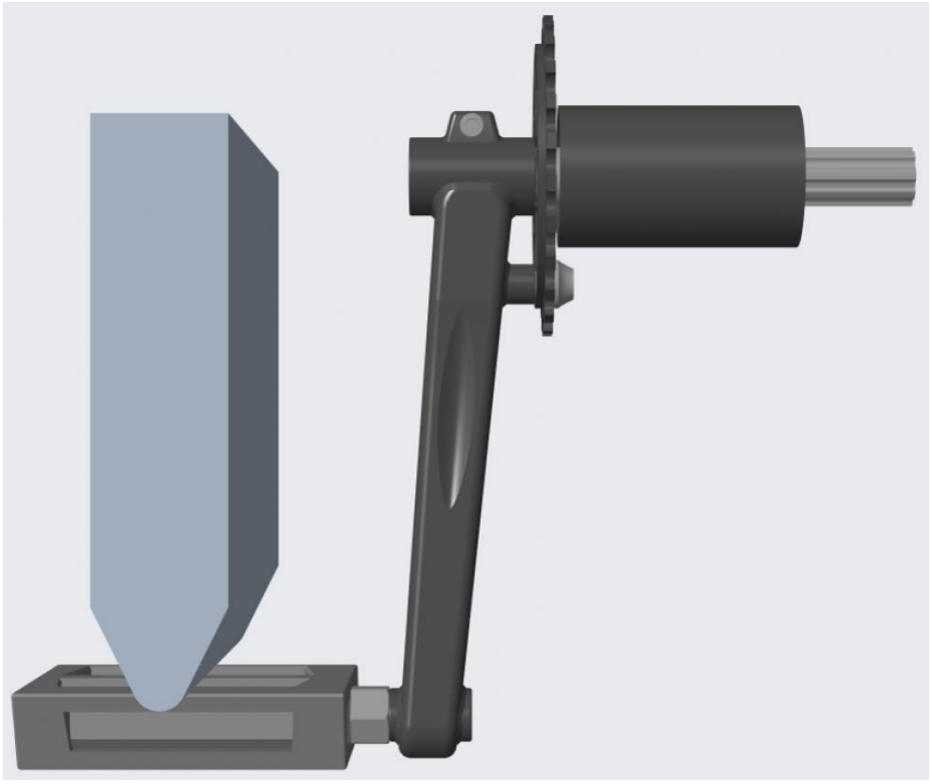


Commercial Bicycle Motorcross (BMX) Crank Assembly

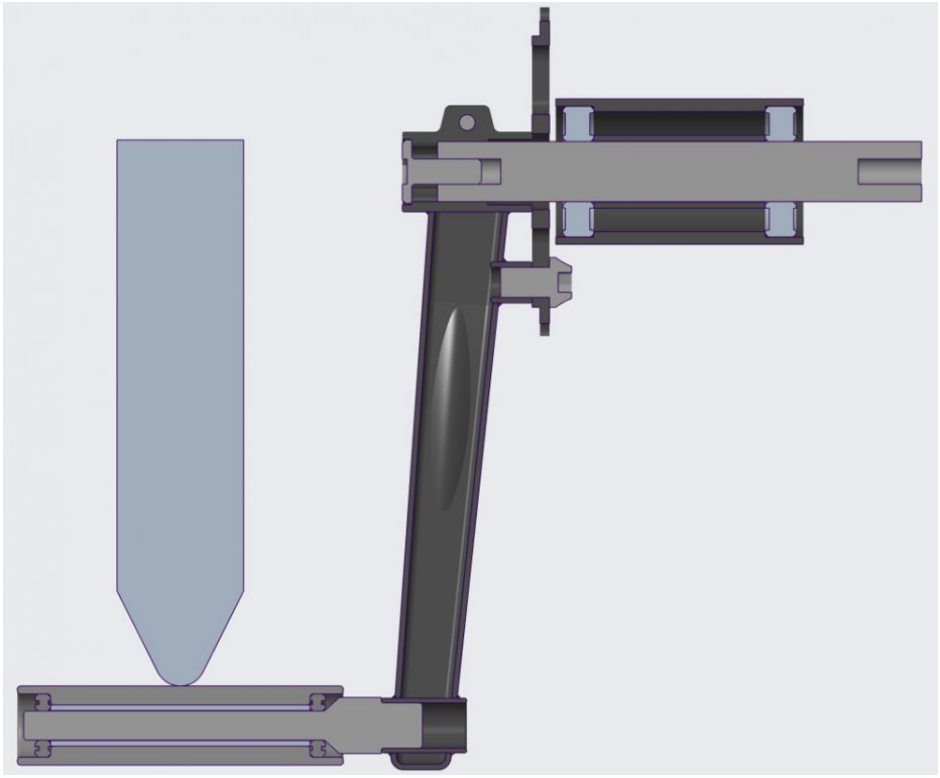
Mechanical Engineering Services:
CAE, Finite Element Analysis, 3D Modeling, CAD

Mechanical Engineering Software:
CREO, ANSYS

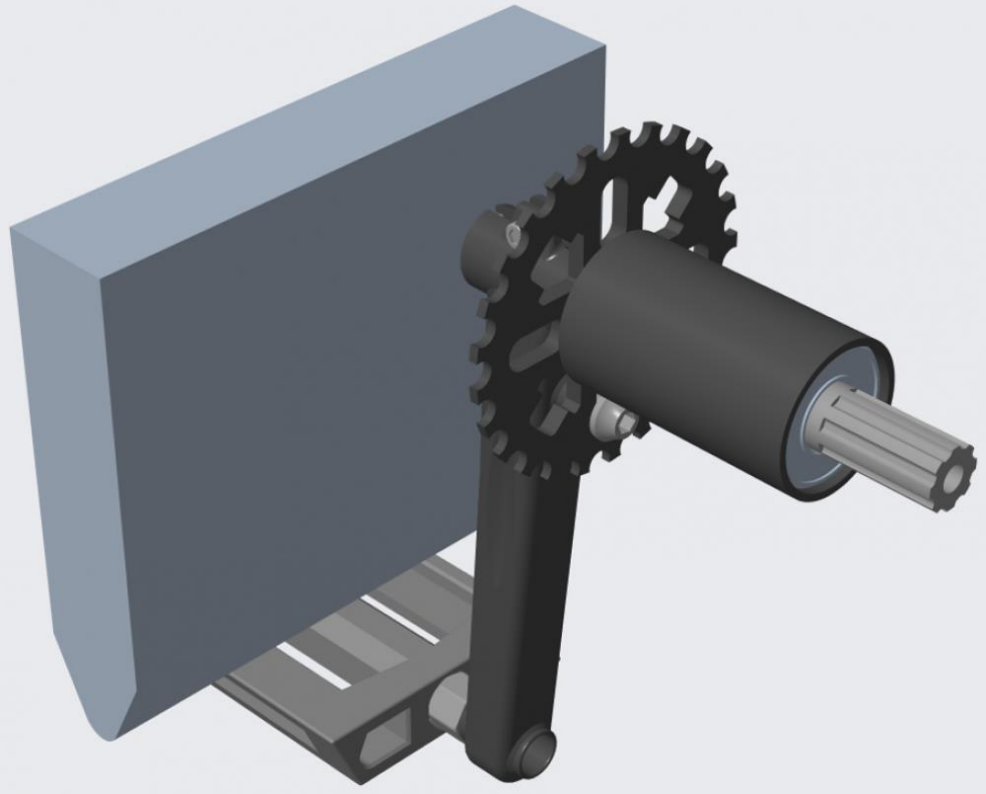
Solution:
The bicycle crank is modeled using computer-aided design (CAD) software. An international BMX performance and safety standard is utilized to implement realistic boundary conditions within the FEA model. Ansys software is then used to obtain the stress and deformation results.



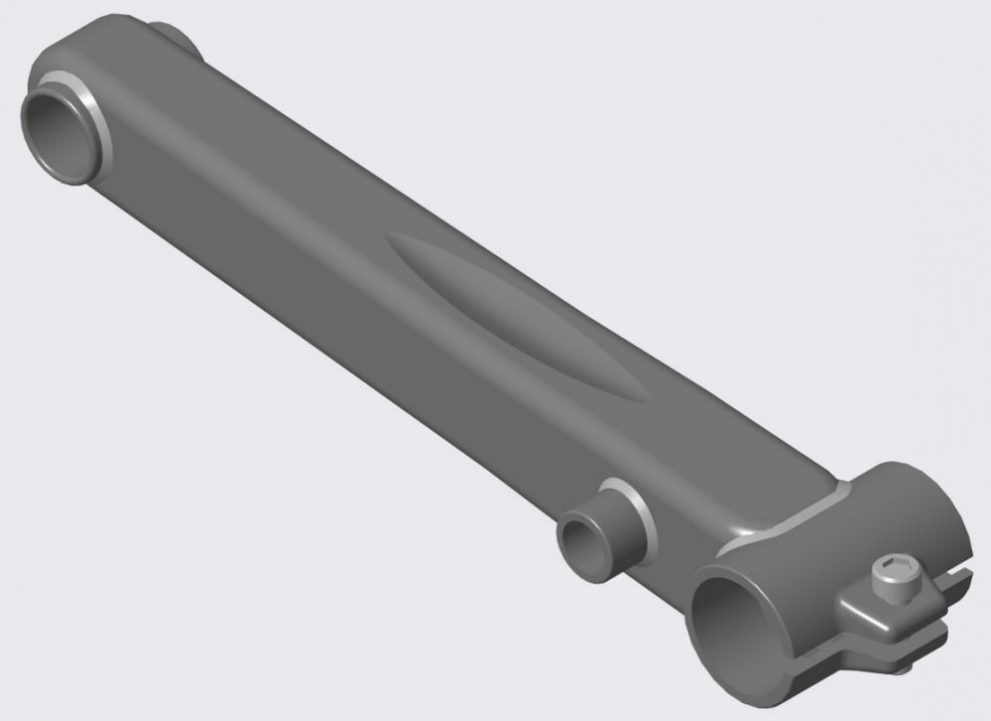
Front View of Bottom Bracket, Crank, Pedal, and Impactor Assembly



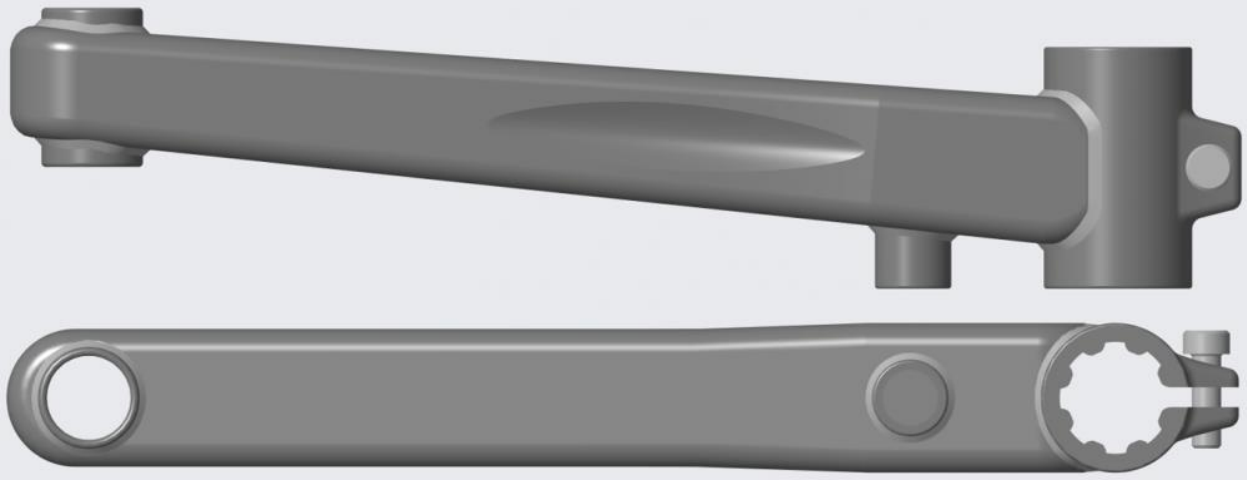
Section View of Bottom Bracket, Crank, Pedal, and Impactor Assembly



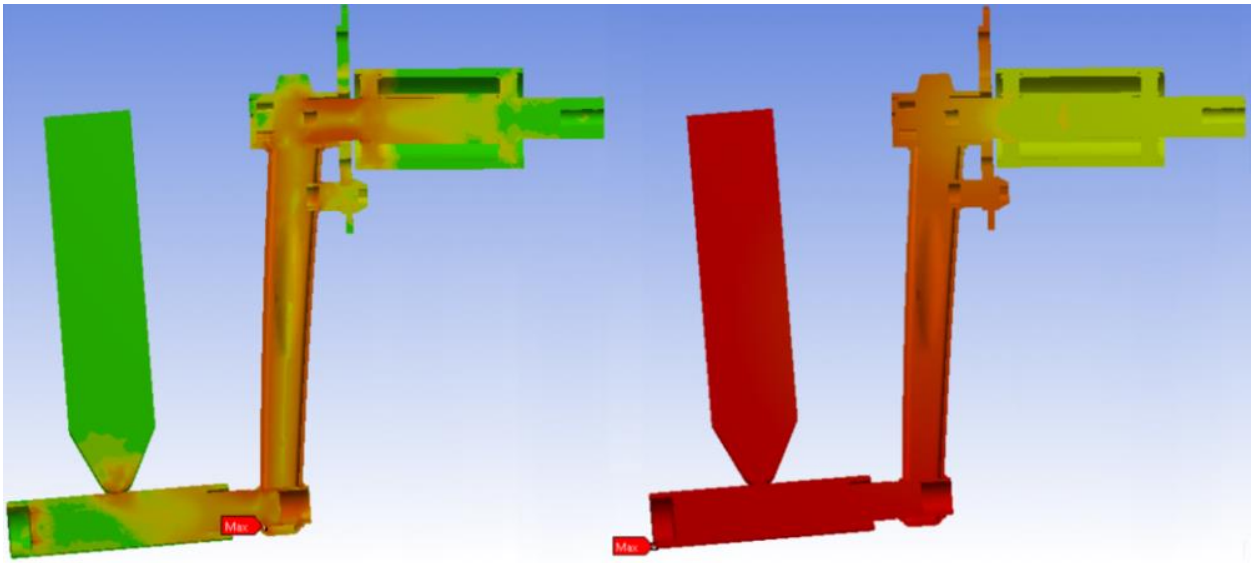
Isometric View of Bottom Bracket, Crank, Pedal, and Impactor Assembly



Isometric View of Crank Assembly



Front and Top View of Crank Assembly



Total Deformation and Von-Mises Equivalent Stress (Logarithmic Scale)

Squat Exercise Machine Redesign

Role:
Graduate student in mechanical engineering design course.

Project goal:
The project task was to redesign a squat exercise machine using dead weights and pulley system to control the mechanism. The original hydraulic weight actuating system needed to be more cost efficient while not sacrificing ergonomics or functionality. The challenge was achieving the necessary mechanical advantage and torque at each joint.

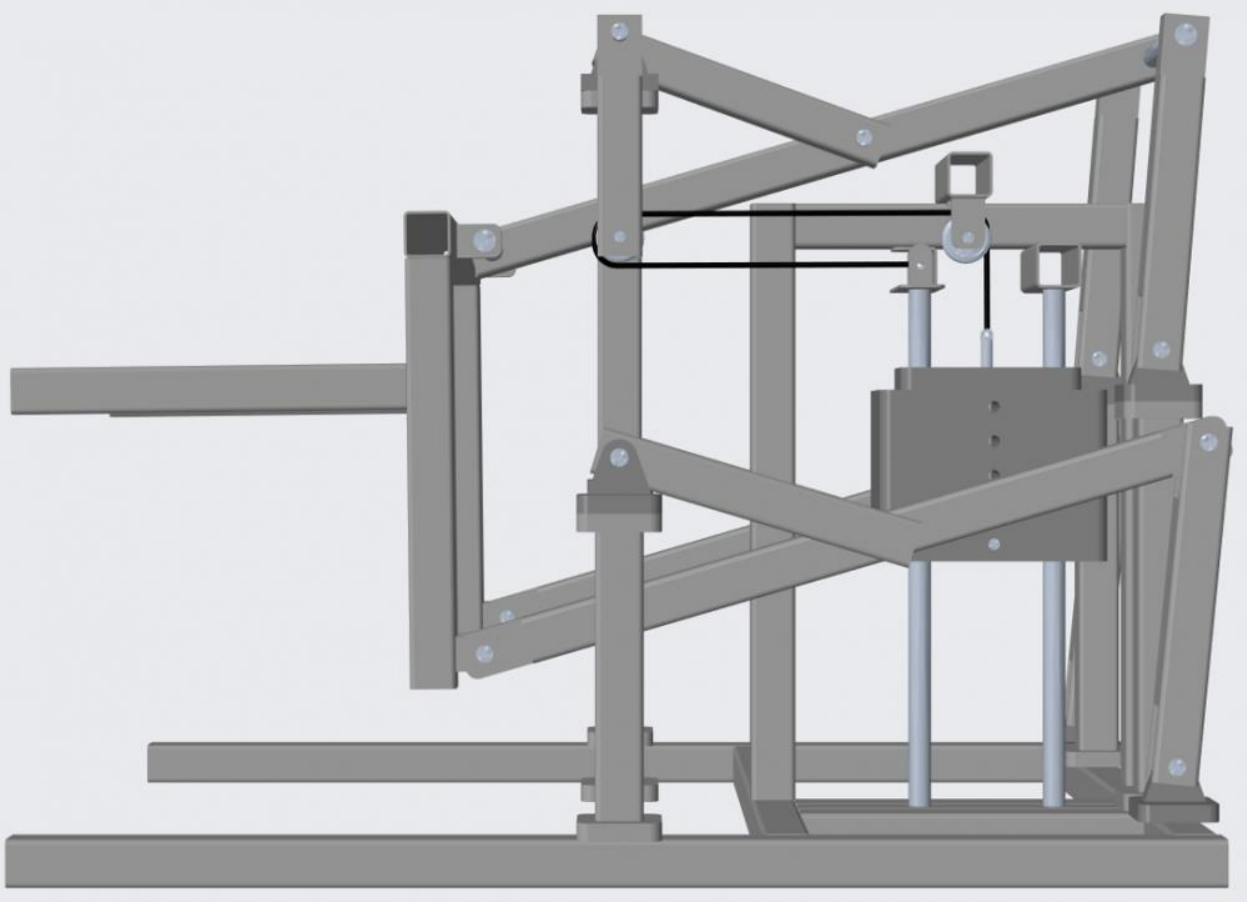


Original Hydraulic Design

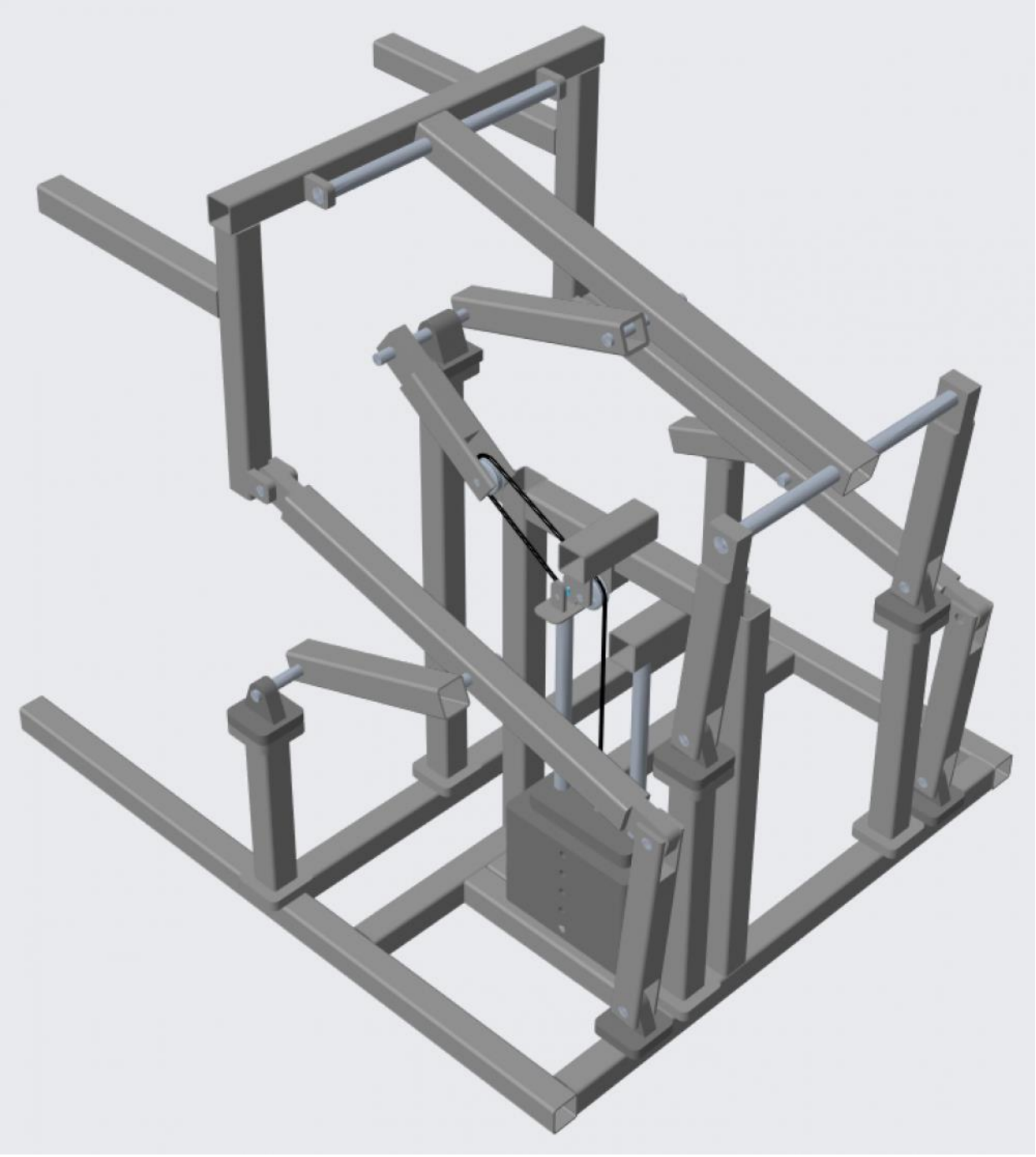
Mechanical Engineering Services:
CAE, Finite Element Analysis, 3D Modeling, CAD

Mechanical Engineering Software:
CREO, ANSYS

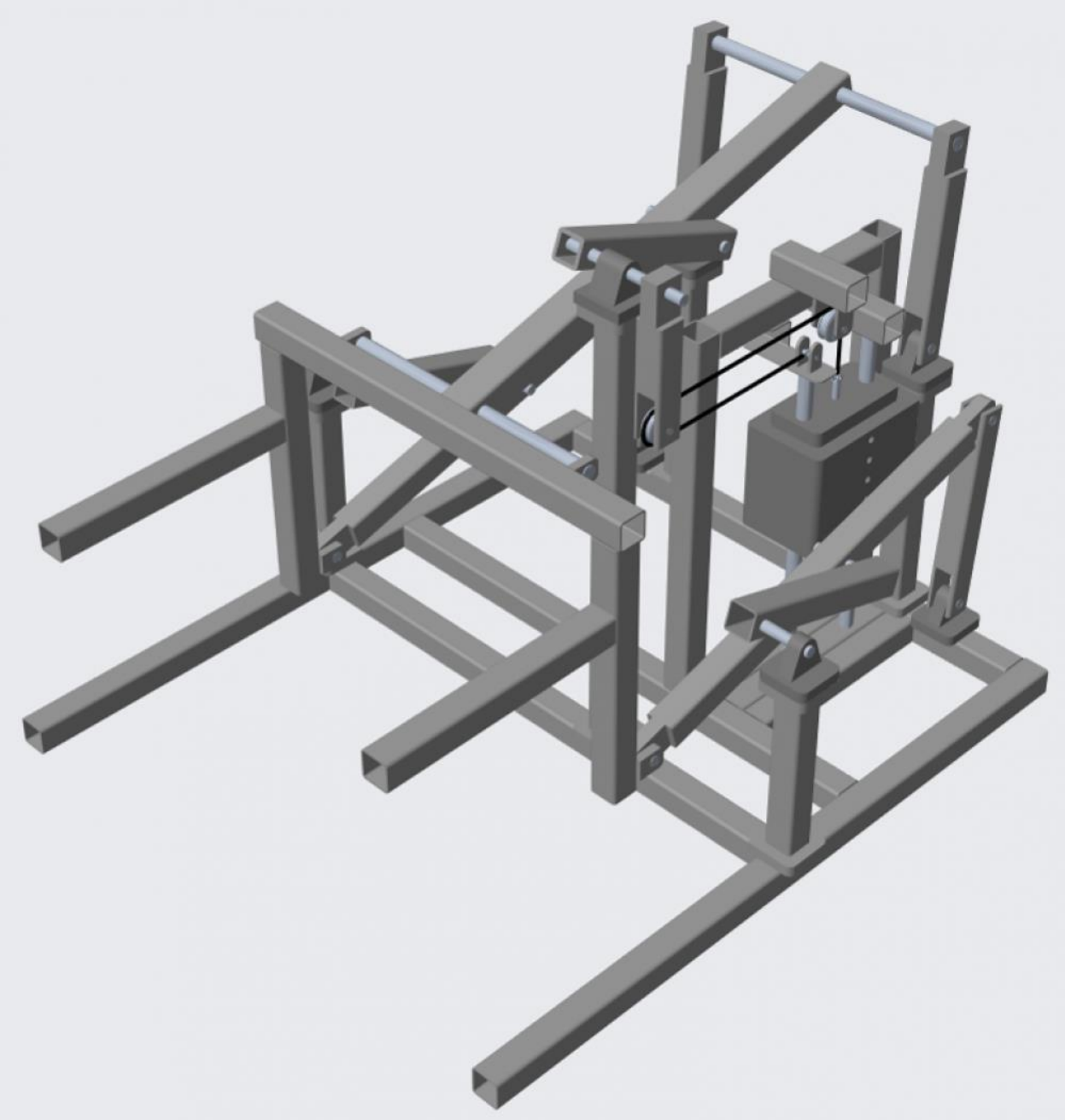
Solution:
I utilized pulleys and steel cables to model a simpler manual weight actuating system. The mechanism was designed using hollow and solid rectangular, square, and circular steel sections. An upper and lower Evans straight line mechanism is utilized. The weights are attached to the intermediate link of the four-bar linkage, which is connected to a pulley system that provides the necessary mechanical advantage. The moment arm length increases as the occupant moves downward to provide lifting assistance. I performed a kinematic and dynamic analysis to validate and visualize the mechanism design.



Front View



Rear Isometric View

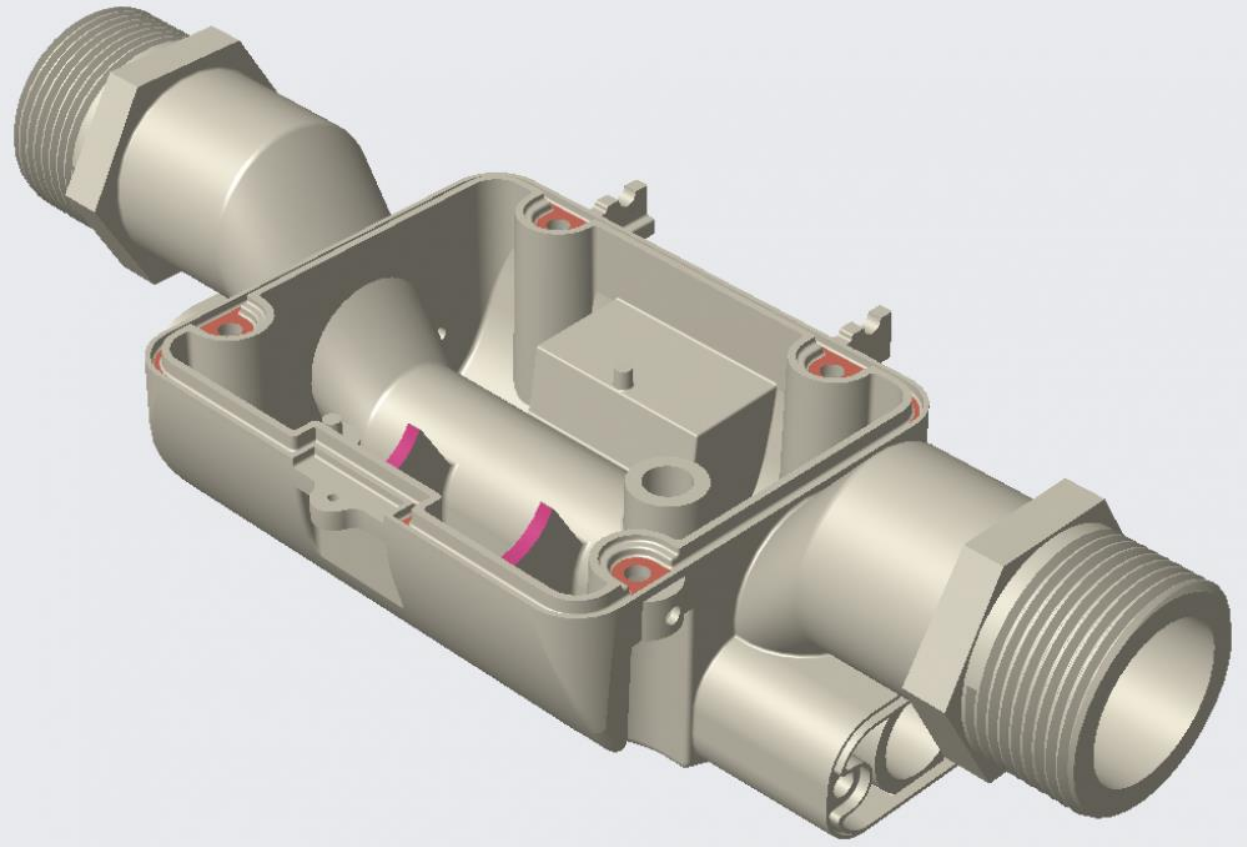


Front Isometric View

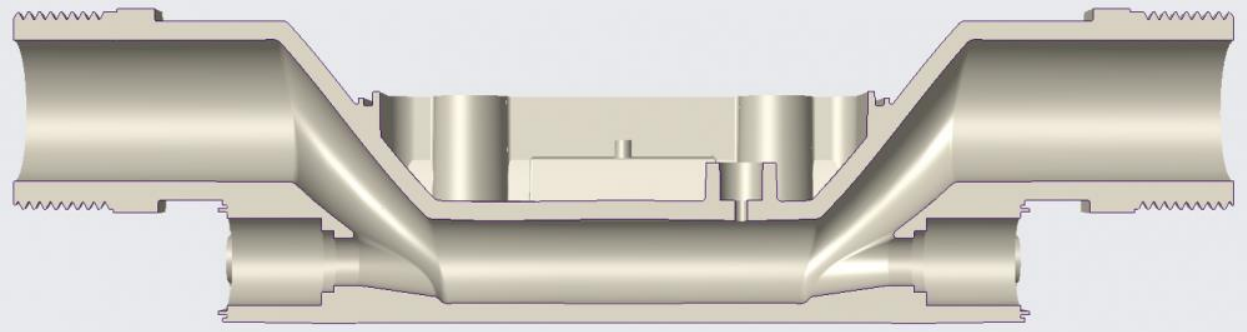
Conjugate Heat Transfer Analysis of Ultrasonic Smart Water Meter

Role:
Mechanical engineer who led the analysis.

Project goal:
The challenge was to determine the flow and heat transfer characteristics of the water meter when subjected to sunlight radiation. The aim was to ensure the thermal expansion did not adversely affect the water meter sensor reading.



Bottom Cover Isometric View

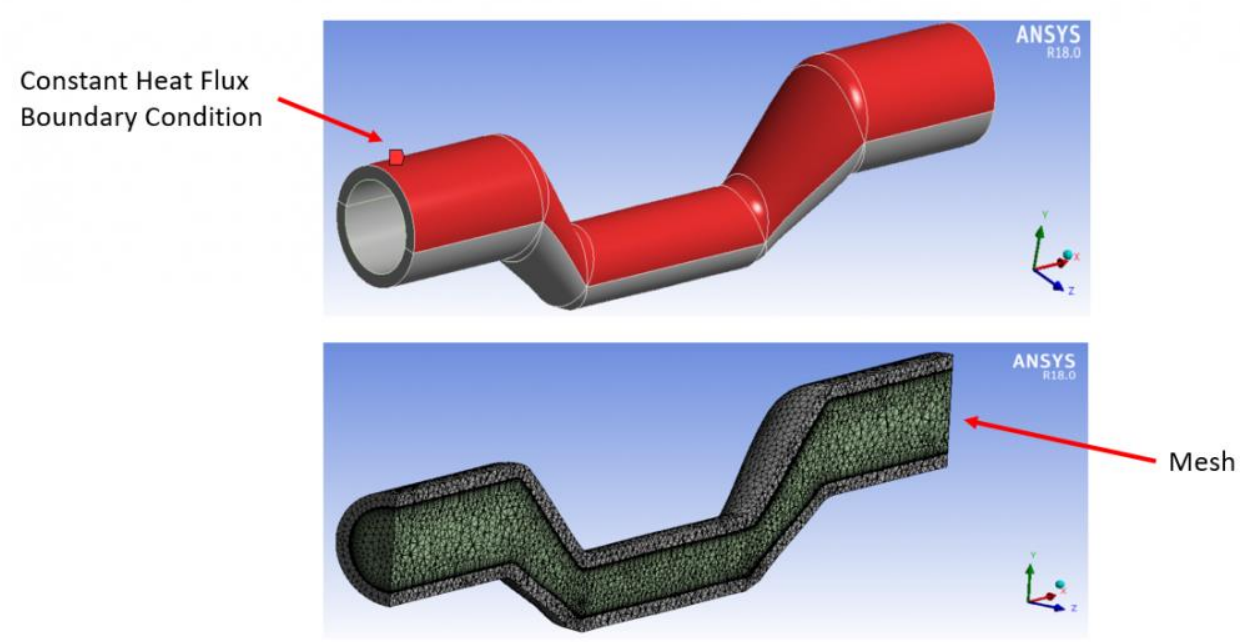


Bottom Cover Section View

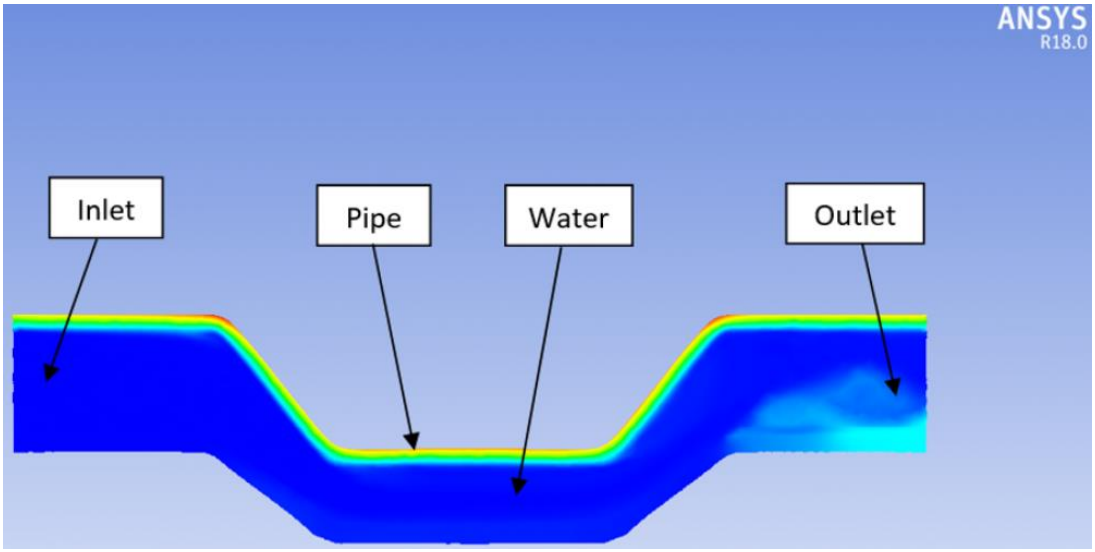
Mechanical Engineering Services:
CAE, Finite Element Analysis, 3D Modeling, CFD Analysis, CAD

Mechanical Engineering Software:
CREO, ANSYS

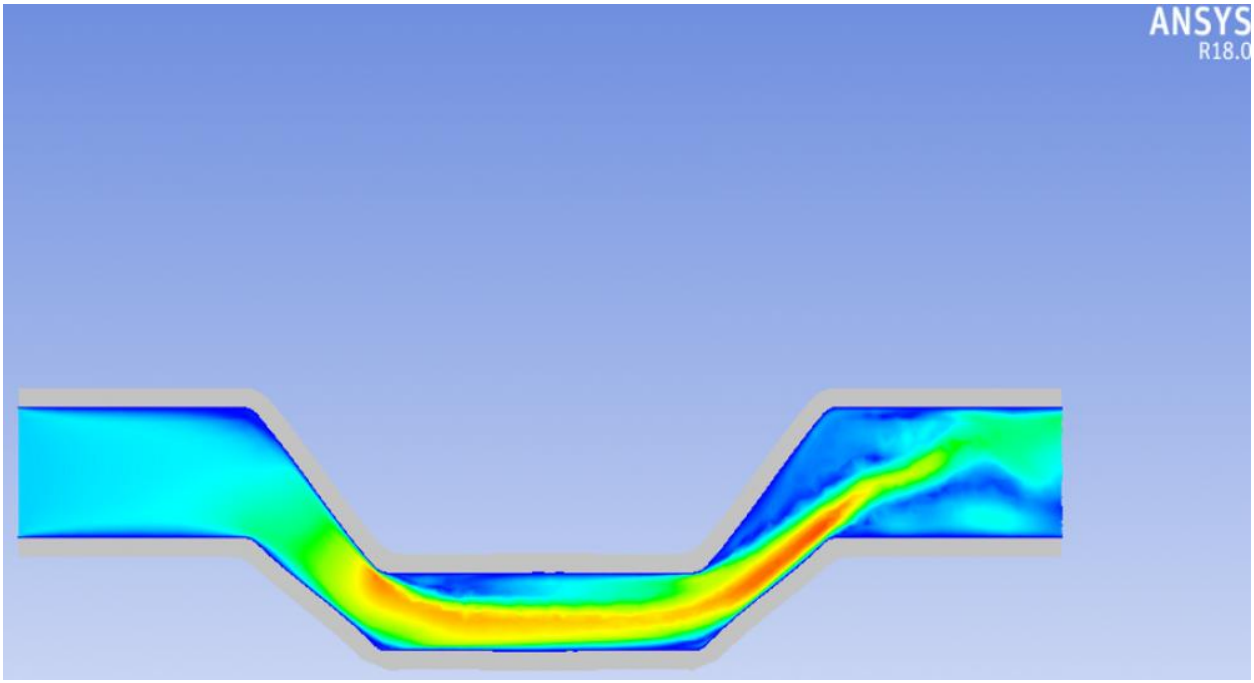
Solution:
The solution involved performing a conjugate heat transfer (CHT) analysis in order to simultaneously obtain the velocity and temperature profile of the water meter pipe. A simplified CAD model of the bottom cover was created to ease the meshing process.



Boundary Conditions and Mesh



Temperature Profile

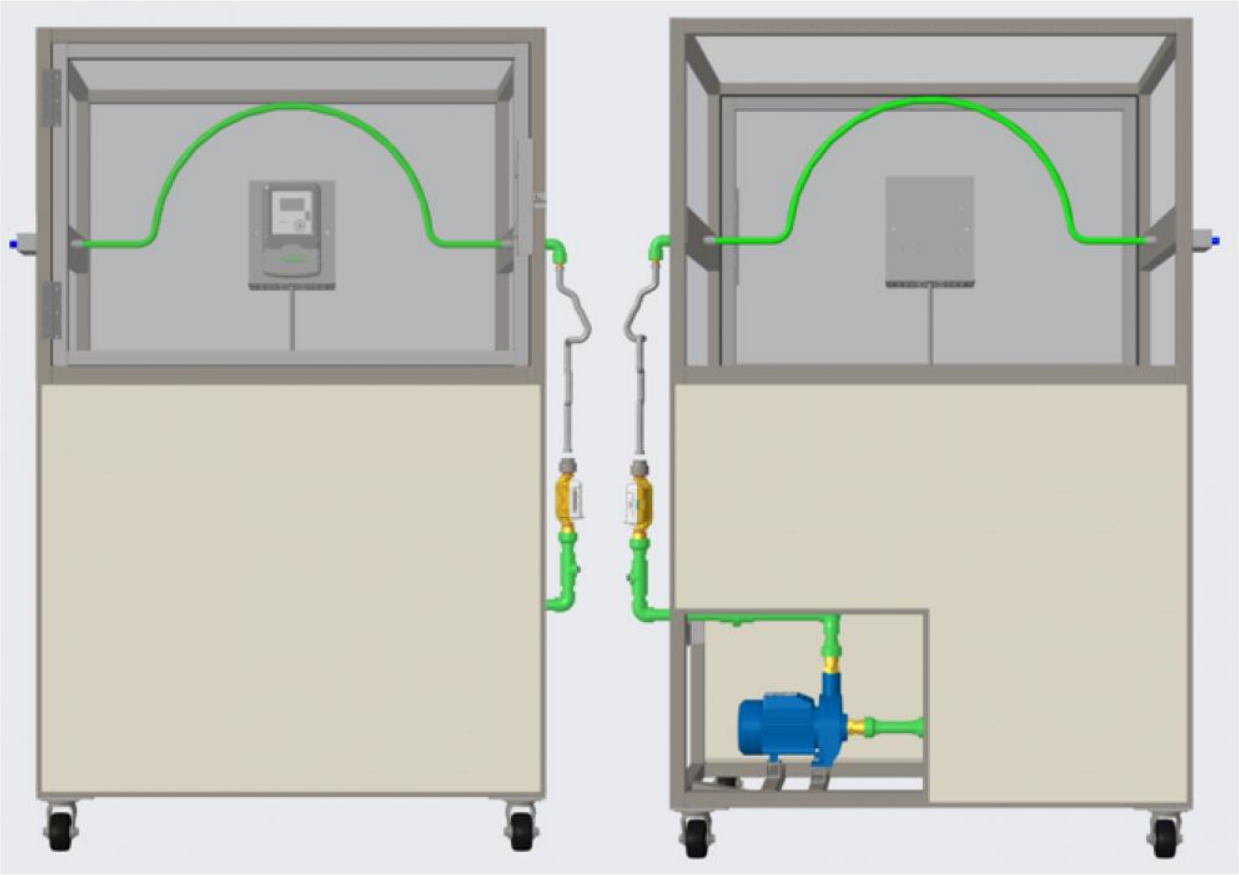


Velocity Profile

Design of Water Chamber for Ingress Protection Testing

Role:
I was the lead mechanical design engineer.

Project goal:
The project involved the new design of an in-house water chamber used to ensure products complied with water ingress protection (IP) standards (X3, X4, and X7). The challenge was to design an automated, portable, corrosion resistant, and leak proof structure that could withstand significant water pressure.

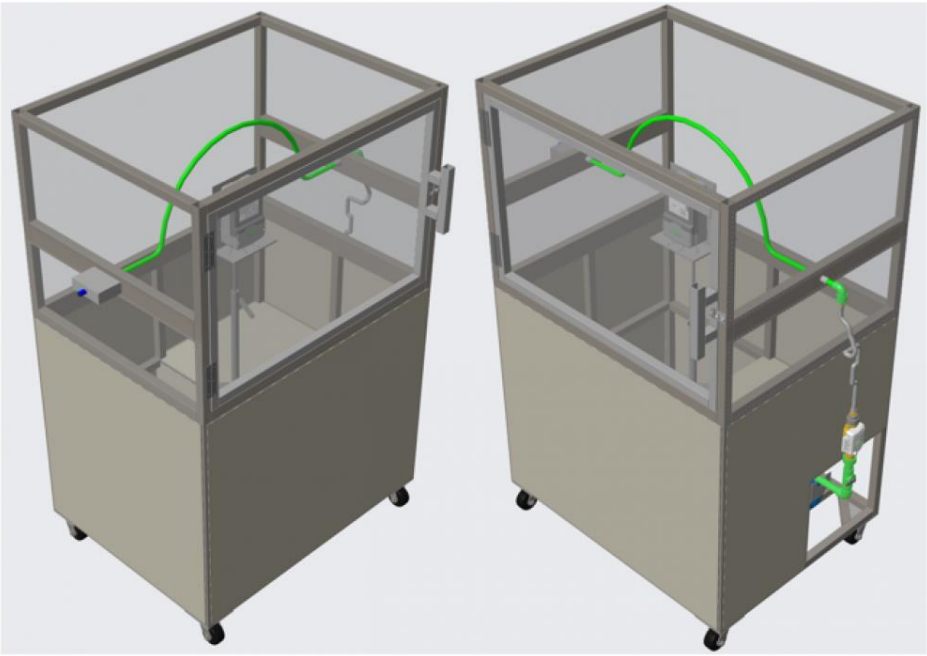


Front and Rear View

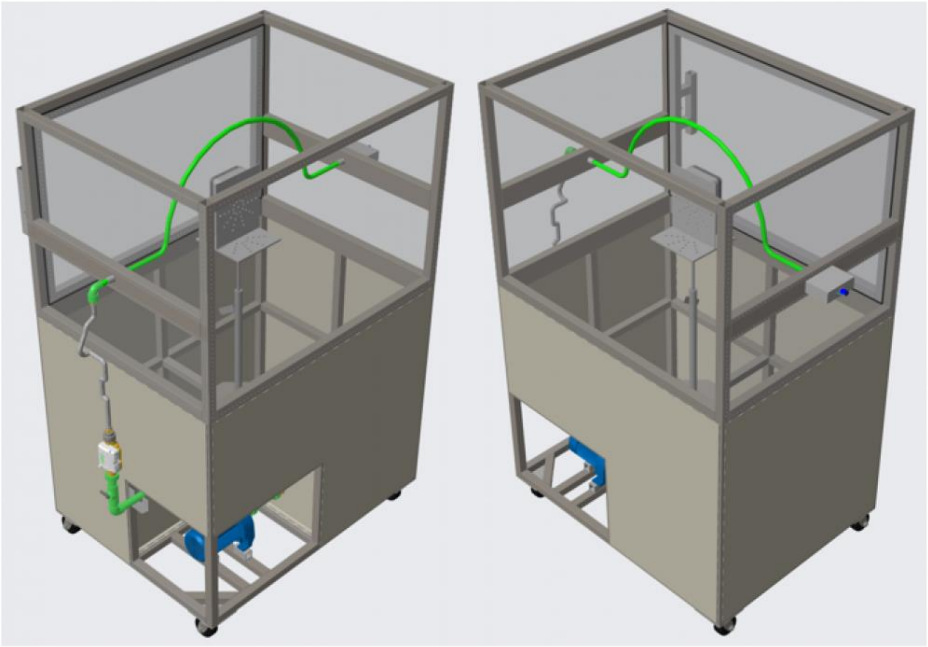
Mechanical Engineering Services:
Product Development, Finite Element Analysis, Design for Manufacturing, 3D Modeling, CAE, CAD

Mechanical Engineering Software:
CREO, ANSYS

Solution:
The chamber was modeled as a weldment structural steel frame composed of rectangular, square, and circular hollow sections in addition to sheet metal coated with corrosion resistant epoxy. The assembly consisted of a stepper motor connected to a plastic oscillating pipe, which sprays the test specimen at a specified pressure and flow rate. The stepper motor is controlled using a micro-controller. The oscillating pipe is attached to a water meter and pump to measure the flow rate and recirculate the water within the chamber.



Front Isometric Views



Rear Isometric Views

Digital Electric Meter Design and Analysis

Role:
I was the lead mechanical design engineer.

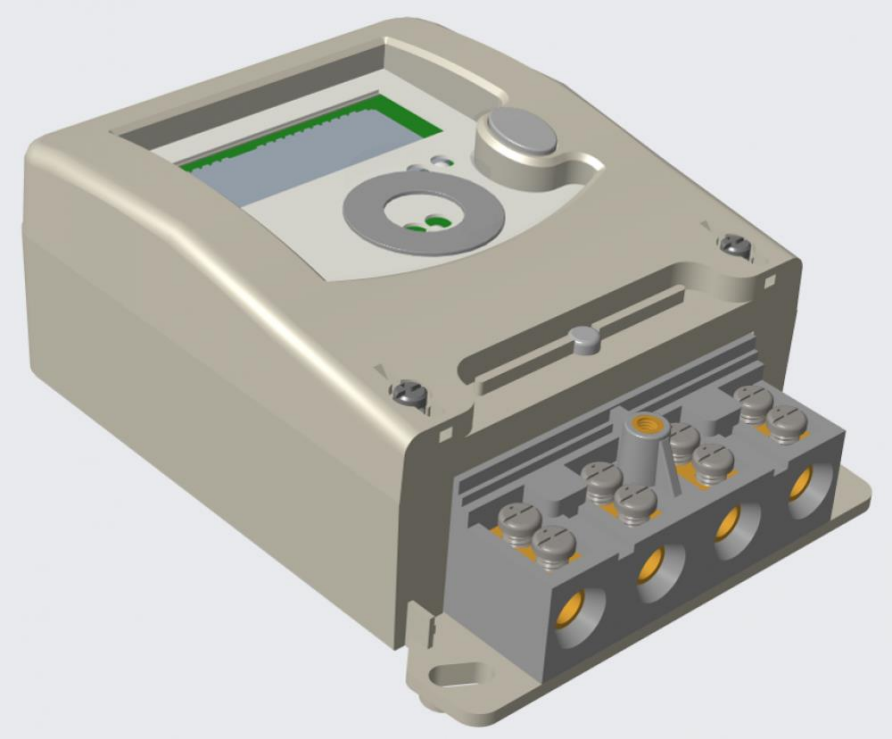
Project goal:
The aim was to design a single phase digital electric meter CAD model assembly. Furthermore, the challenge was to predict the meter's impact resistance and internal temperature under maximum voltage and amperage conditions.



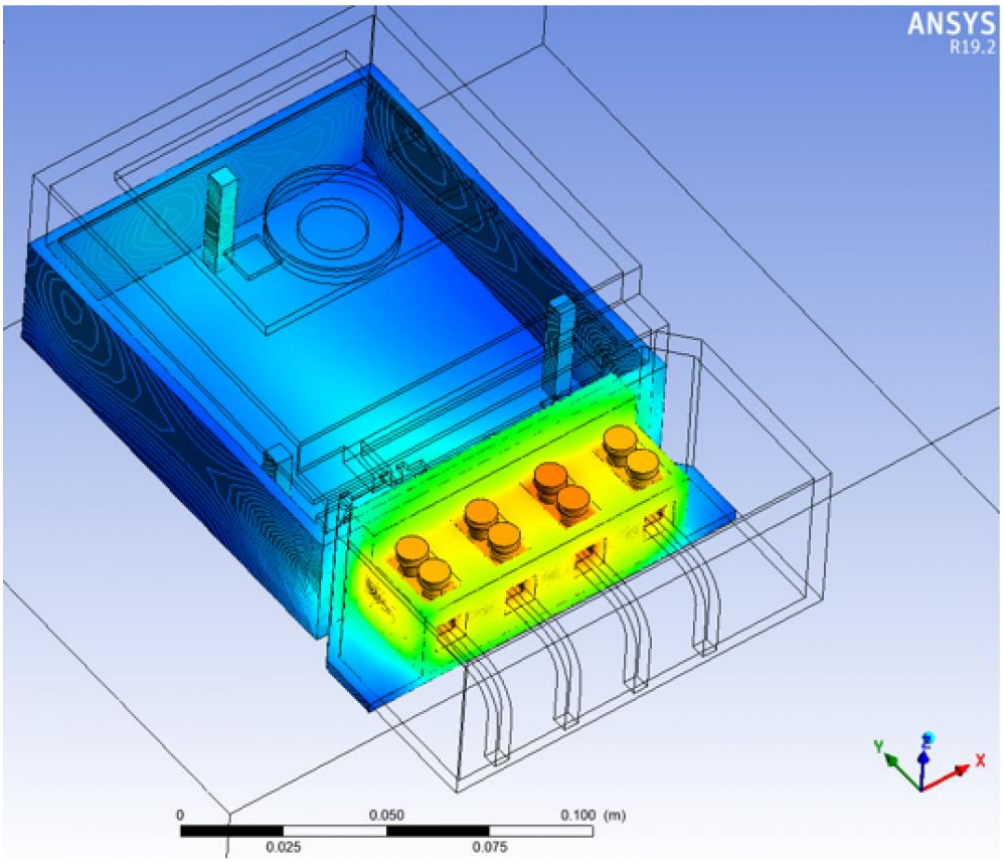
Mechanical Engineering Services:
Reverse Engineering, Product Development, Finite Element Analysis, Design for Manufacturing, 3D Modeling, CAE, CAD

Mechanical Engineering Software:
CREO, ANSYS

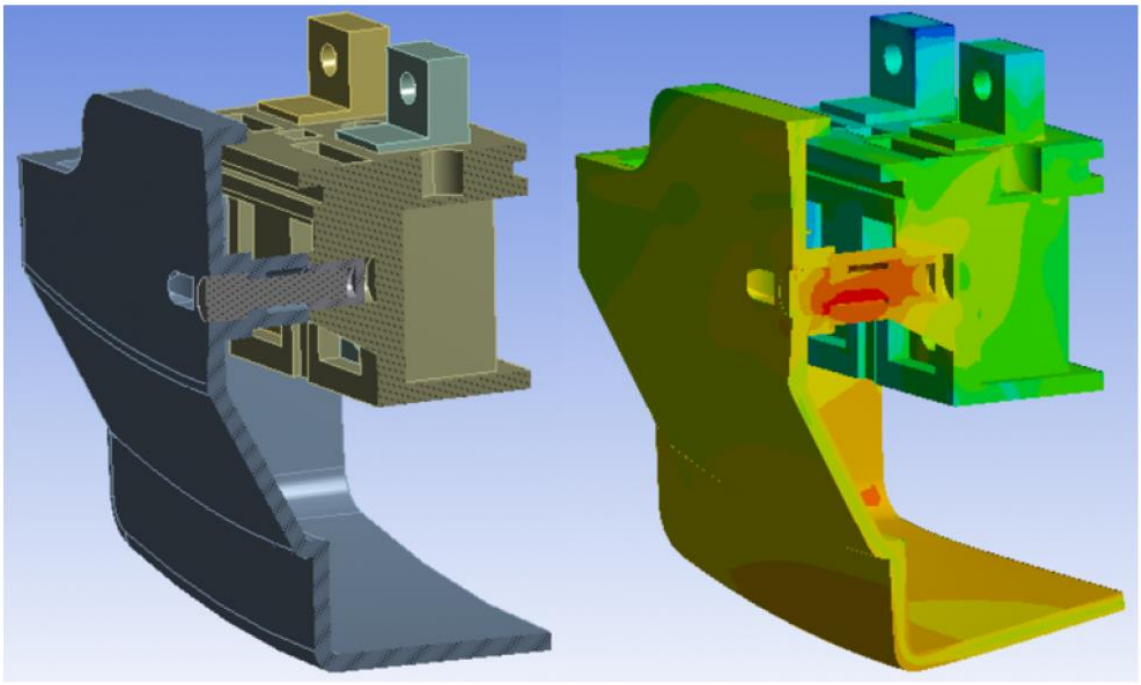
Solution:
I designed the 3D CAD model assembly and detailed 2D mechanical drawings using Creo Parametric modeling software. Also performed impact/drop and natural convection heat transfer simulations, which were validated through experimental tests.







Natural Convection Heat Transfer Simulation

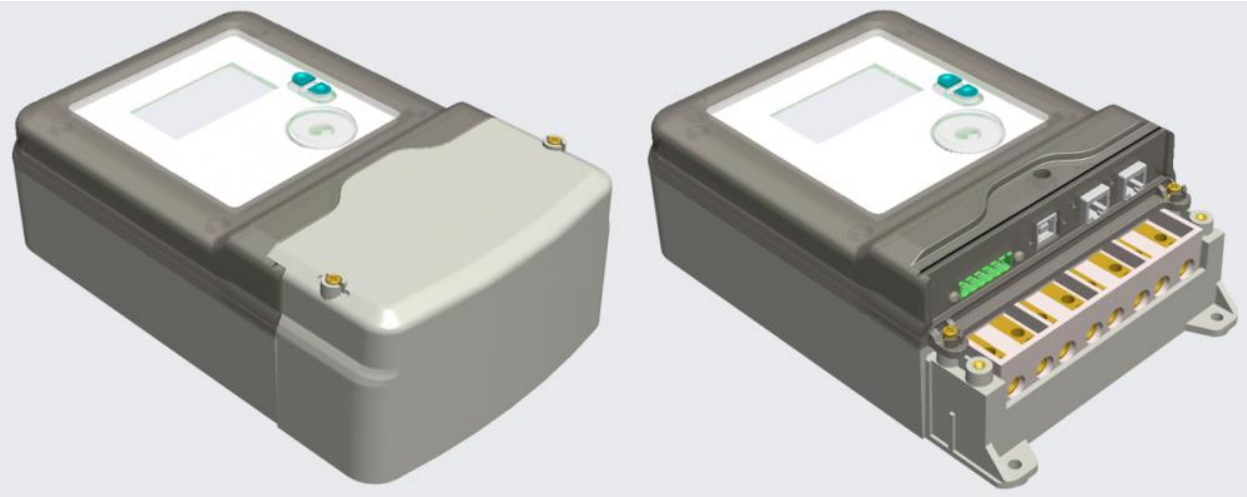


Bottom Cover and Terminal Block Impact/Drop Finite Element Analysis

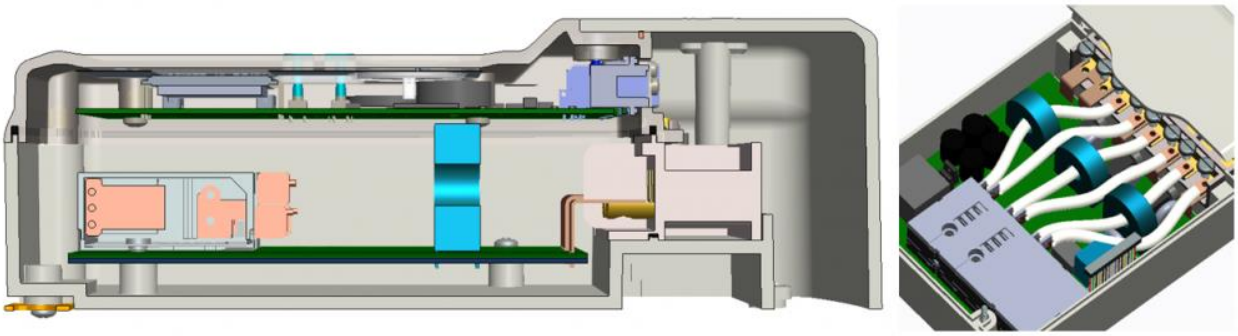
Heat Transfer Simulation of Three Phase Electric Meter

Role:
Mechanical engineer who led simulations and testing.

Project goal:
The challenge was to predict the temperature within the utility meter under maximum voltage and amperage conditions.



Electric Meter Assembly Isometric View

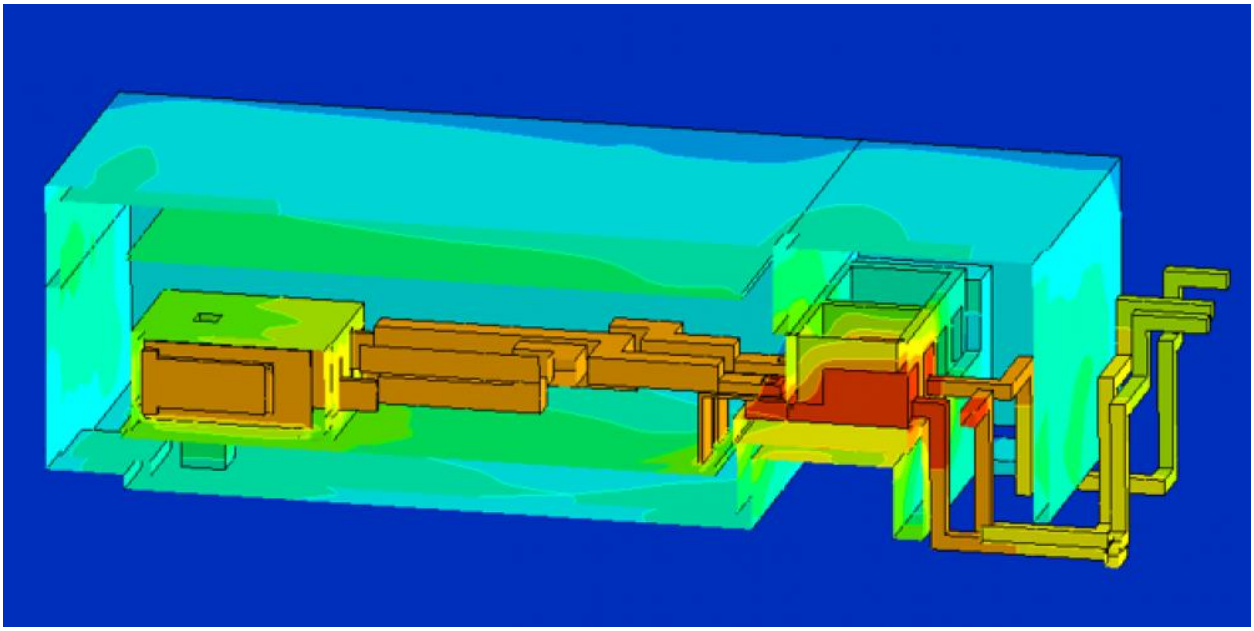


Electric Meter Assembly Internal/Section View

Mechanical Engineering Services:
CFD Analysis, CAD

Mechanical Engineering Software:
CREO, ANSYS

Solution:
Modeled simplified 3D CAD model assembly of the meter and performed natural convection heat transfer simulations, which were validated through experimental tests.

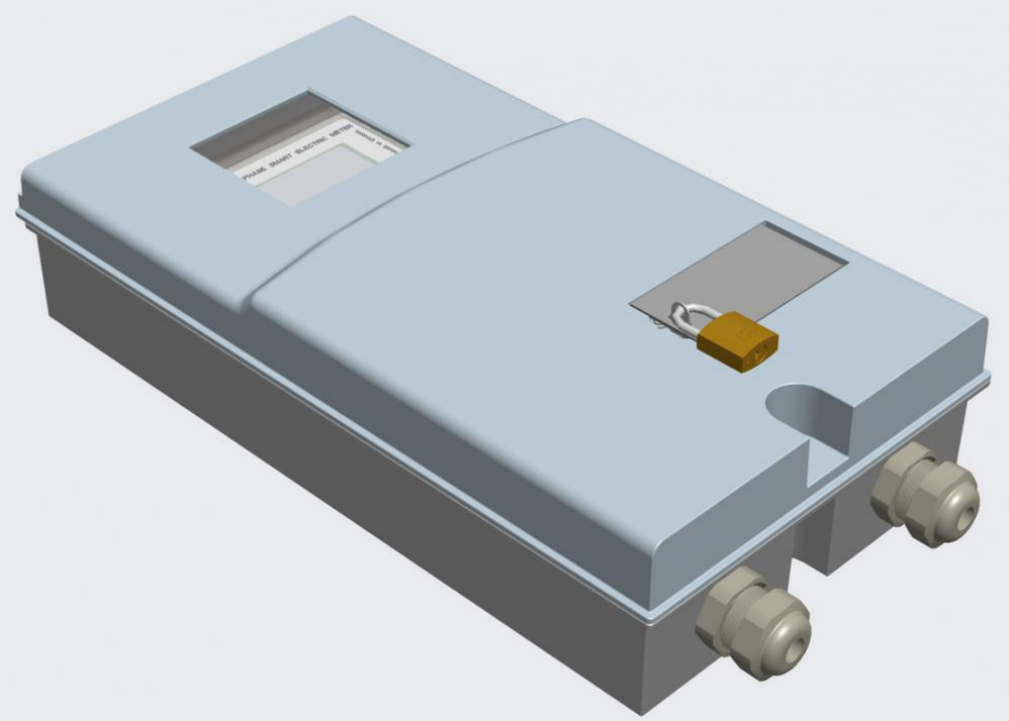


Temperature Profile (Section View)

Electronic Enclosure Design

Role:
I was the lead mechanical design engineer.

Project goal:
The project task was to design an electronic enclosure with IP 54 level water and dust ingress protection. Additionally, the enclosure needed to withstand hot/cold temperatures, sunlight radiation, impacts/drops, fire, theft, and be able to house both single and three phase electric utility meters.

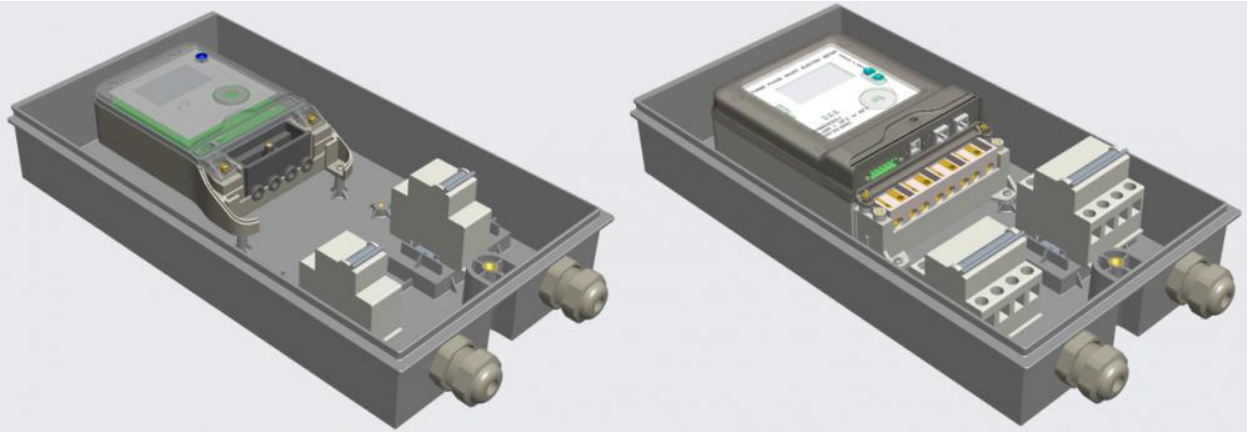


Electronic Enclosure Assembly Front View

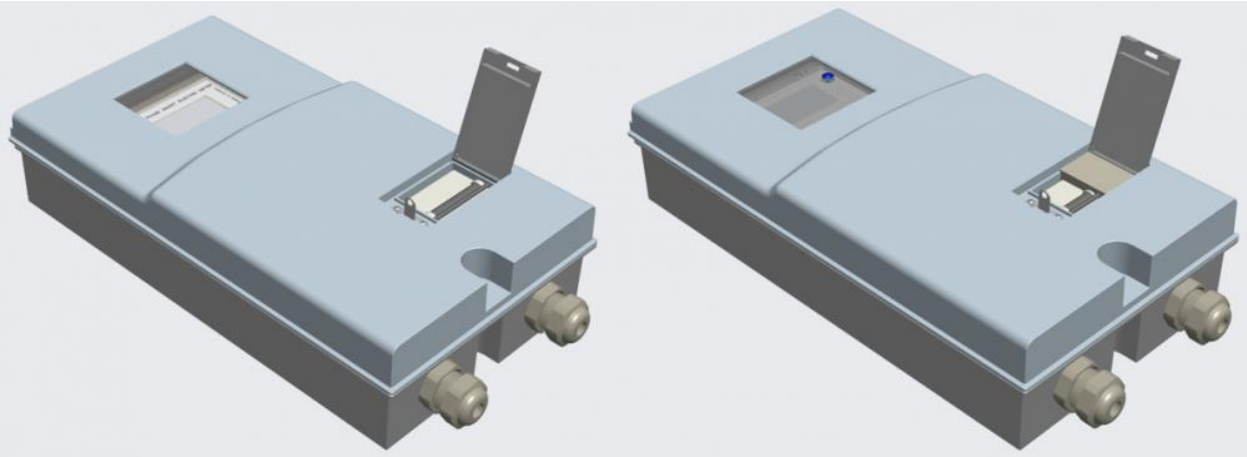
Mechanical Engineering Services:
Reverse Engineering, Product Development, Finite Element Analysis, Design for Manufacturing, 3D Modeling, CFD Analysis, CAE, CAD

Mechanical Engineering Software:
CREO, ANSYS

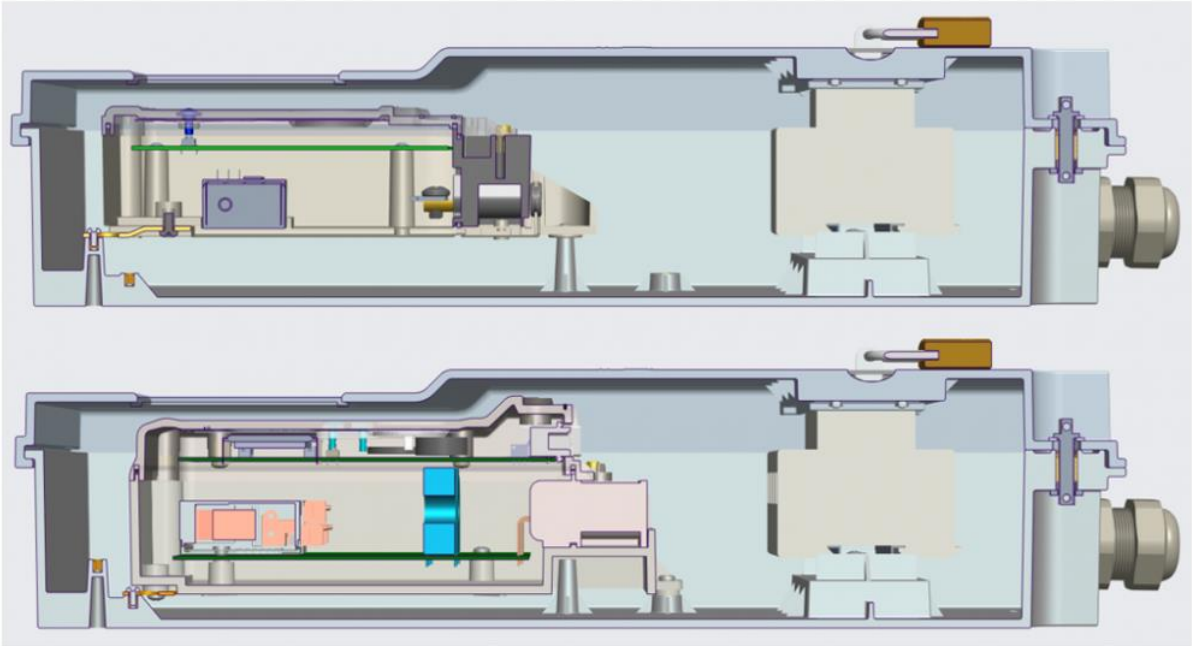
Solution:
The project solution involved designing a compact and light weight plastic enclosure equipped with bosses and DIN rails for mounting multiple utility meters and circuit breakers. Additionally, tamper proof padlocks and steel sealing wires are utilized along with IP 54 level cable glands.



Internal Isometric View



External Isometric View



Section View

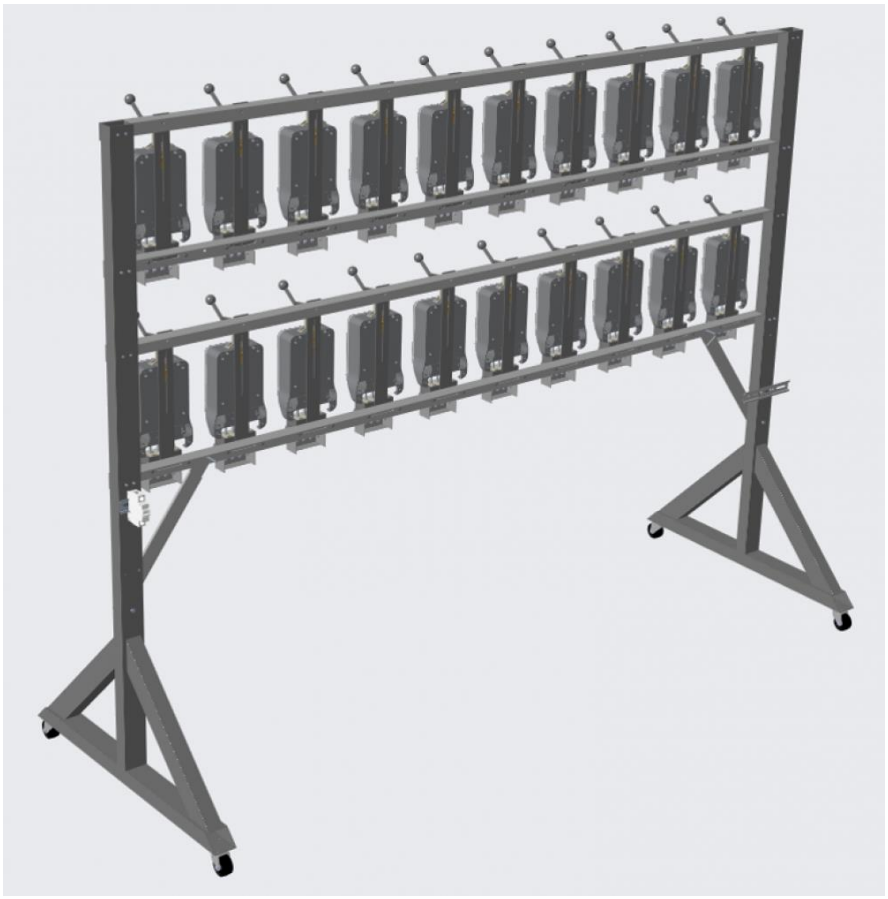
Power Line Communications Test Stand Design

Role:
I was the lead mechanical design engineer.

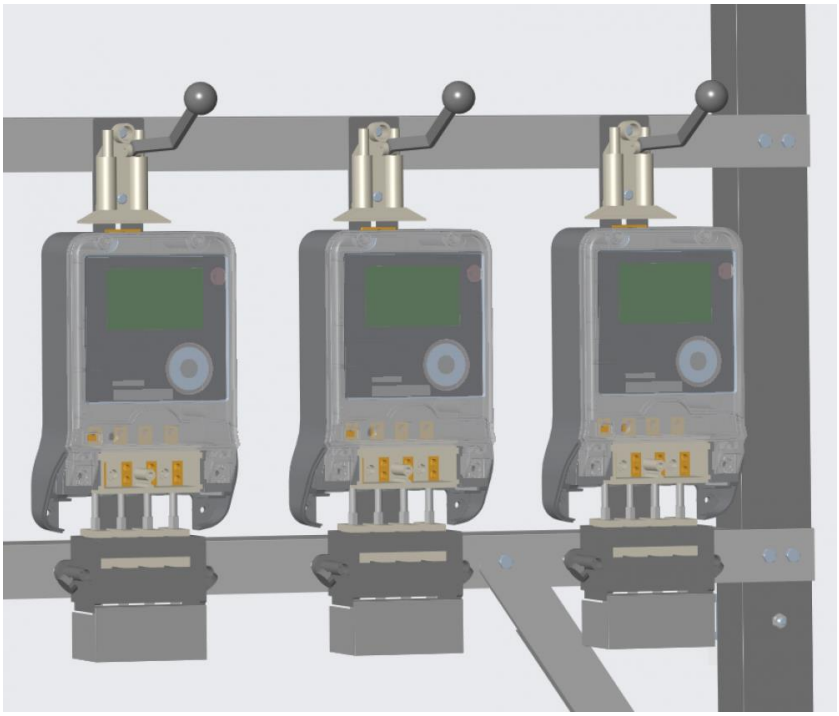
Project goal:
The task was to design a power line communications test stand fixture in order to perform in-house testing of electric meters operating under low current and voltage conditions.



Front Isometric View



Rear Isometric View

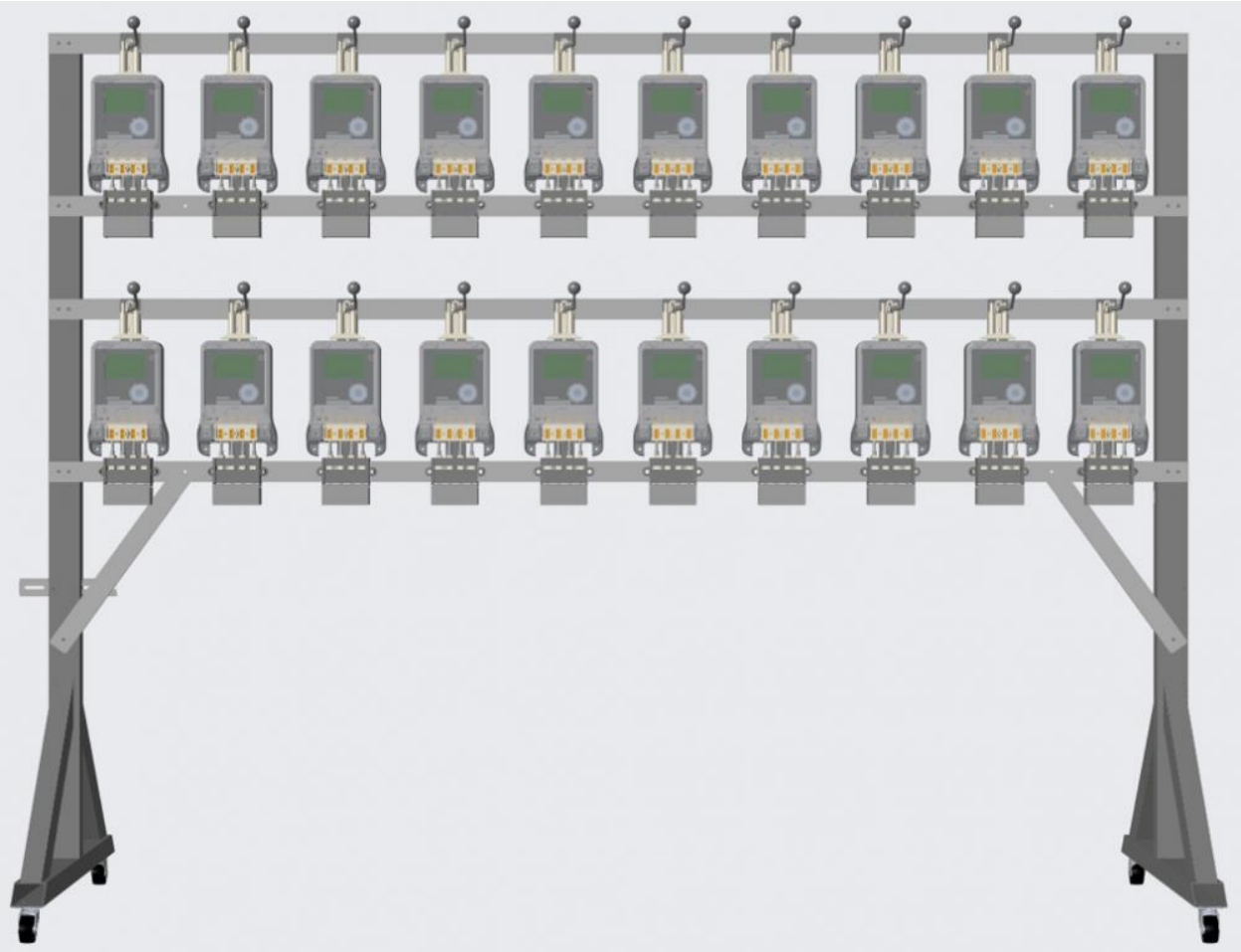


Close-Up View

Mechanical Engineering Services:
Reverse Engineering, Product Development, Finite Element Analysis, Design for Manufacturing, 3D Modeling, CFD Analysis, CAE, CAD

Mechanical Engineering Software:
CREO, ANSYS

Solution:
Designed a 3D CAD model assembly of the PLC stand using rectangular and square steel hollow sections, steel angle, toggle clamps, and mounting pins equipped with plastic safety covers.

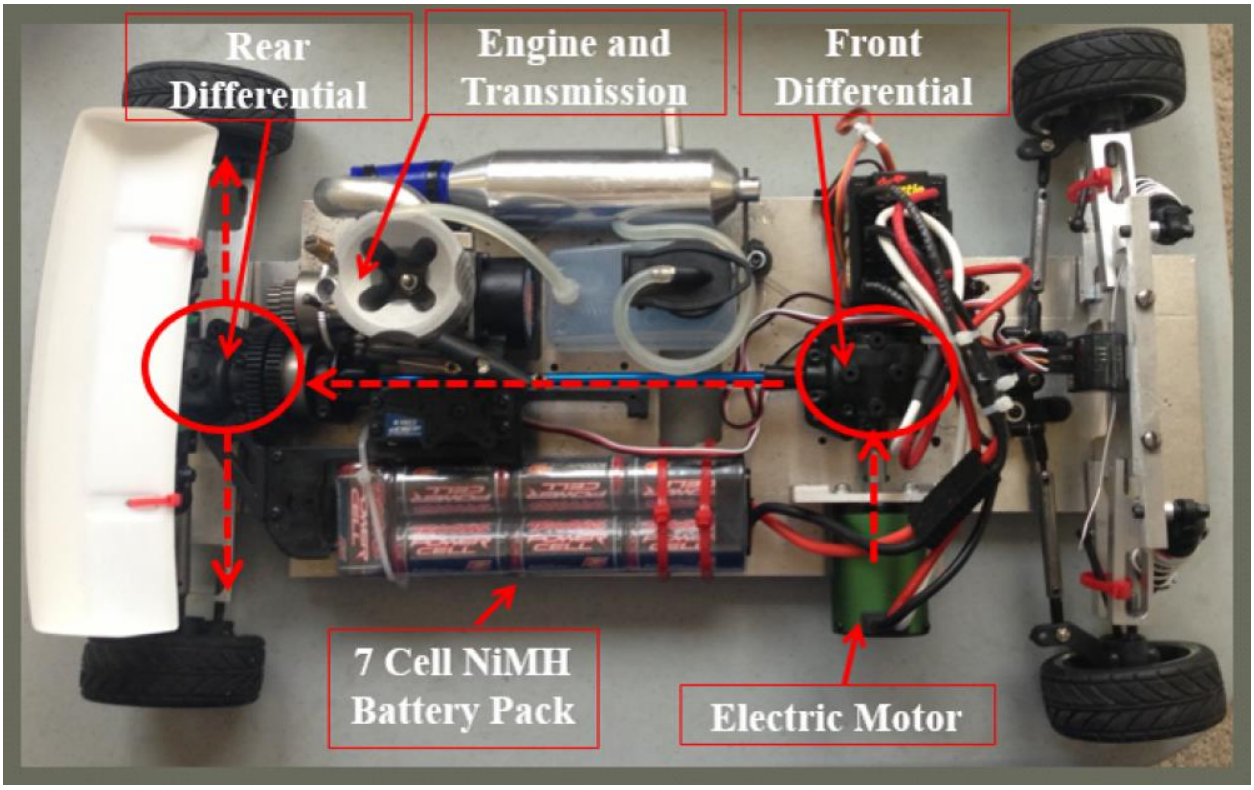


Front View

1/8th Scale Parallel Hybrid Remote Controlled Car

Role:
Mechanical engineering student within group senior design project.

Project goal:
The project is the design and build of a 1/8th scale parallel hybrid remote controlled (RC) car. We utilized components from a commercial RC car and fabricated other parts for the suspension system and frame.

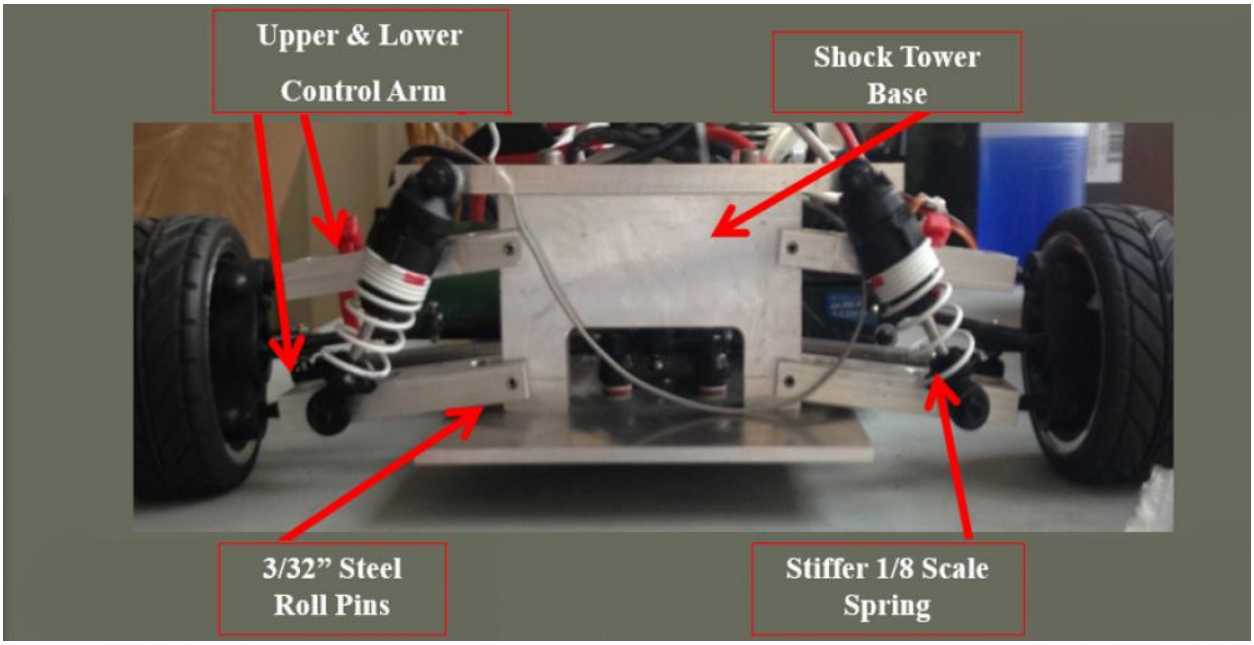


RC Car Assembly Top View

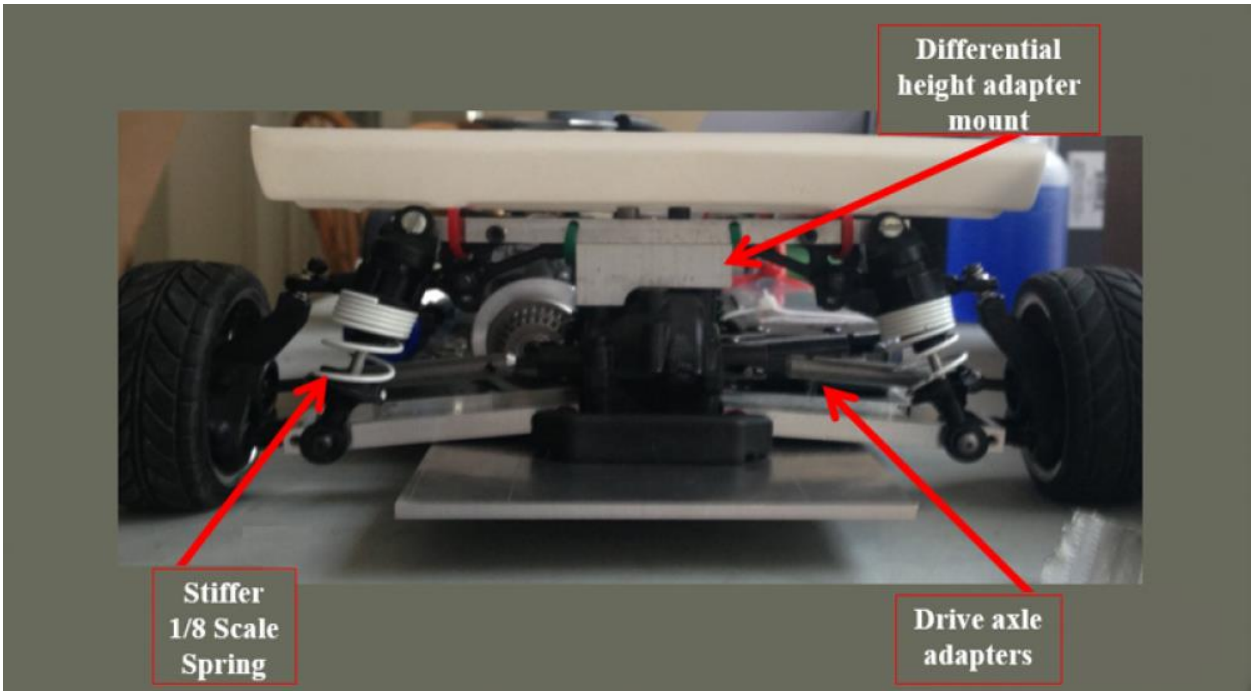
Mechanical Engineering Services:
Reverse Engineering, Product Development, Finite Element Analysis, Design for Manufacturing, 3D Modeling, CAE, CAD

Mechanical Engineering Software:
Solidworks, MSC Nastran/Patran

Solution:
The RC car is made from machined parts including an aluminum chassis plate, shock tower, and control arms that make up the suspension system. The assembly includes an electric motor powered by NiMH batteries and a nitromethane single cylinder engine.



Front Suspension System



Rear Suspension System